

WILLIAM KIRSCHENMAN

Ph.D. Candidate, Department of Industrial and Systems Engineering, North Carolina State University

115 Hilda Grace Lane ♦ Cary, NC 27519

(910) · 527 · 5377 ♦ wkkirsch@ncsu.edu

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EDUCATION

Ph.D. (<i>In progress</i>), Operations Research, <i>North Carolina State University</i>	Anticipated <i>May 2026</i>
M.S. , Operations Research, <i>George Mason University</i>	<i>2020</i>
MPA , Public Administration, <i>Webster University</i>	<i>2015</i>
B.S. , Mechanical Engineering, <i>United States Military Academy</i>	<i>2010</i>

FELLOWSHIPS AND AWARDS

Salah E. Elmaghraby Distinguished Student Award, Operations Research Program, NC State University	<i>2025</i>
General Omar N. Bradley Research Fellowship in Mathematics	<i>2024, 2025</i>
Army Operations Research Symposium (AORS) Best Presentation, Wargaming Working Group	<i>2021</i>
Outstanding Academic Achievement Award in Operations Research, George Mason University	<i>2020</i>

PUBLICATIONS

JOURNAL ARTICLES IN PEER-REVIEW

Kirschenman, W., Heese, H.S., Kay, M., King, R., McConnell, B. (2026). Enhancing Military Load Planning: A Prioritized 2-D Orthogonal Packing Approach, In review at *Omega*.

Kirschenman, W., Heese, H.S., Kay, M., King, R., McConnell, B. (2025). The 2-D Orthogonal Packing Problem with Multiple Levels of Prioritization: A Spatial Optimization Perspective, Under major revision at *Naval Research Logistics*.

TRADE JOURNAL PUBLICATIONS (NON-REFEREED)

Kirschenman, W., McConnell, B., King, R. (2024). Automated Vessel Selection and Combat Load Planning. *Army Sustainment*, Vol 56, No 4, 58-61. ([Link](#))

PRESENTATIONS

Kirschenman, W., Heese, H.S., Kay, M., King, R., McConnell, B. (2025). “A Prioritized 2-D Packing Model for Military Combat Loading.” Presentation at the Annual Meeting of the Institute for Operations Research and the Management Sciences (INFORMS), Atlanta, GA. ([Link](#))

Kirschenman, W., Heese, H.S., Kay, M., King, R., McConnell, B. (2025). “Automating Combat Load Planning using a Prioritized 2-D Packing Approach.” Presentation at the 2025 Army Operations Research Symposium (AORS) Symposium, Aberdeen Proving Ground, MD. ([Link](#))

Kirschenman, W., Heese, H.S., Kay, M., King, R., McConnell, B. (2025). “The 2-D Bin Packing Problem with Multiple Levels of Prioritization: A Spatial Optimization Perspective.” Presentation at the 93rd Military Operations Research Society (MORS) Symposium, National Conference Center, Leesburg, VA. ([Link](#))

Kirschenman, W., Heese, H.S., Kay, M., King, R., McConnell, B. (2025). “Automating Military Load Planning: A Prioritized 2-D Bin Packing Approach.” Presentation at the 93rd Military Operations Research Society (MORS) Symposium, National Conference Center, Leesburg, VA. ([Link](#))

Kirschenman, W., Heese, H.S., King, R., McConnell, B. (2025). “The 2-D Bin Packing Problem with Multiple Levels of Prioritization: A Spatial Optimization Perspective.” Poster presentation, Graduate Student Research Symposium, NC State University, Raleigh, NC. ([Link](#))

Kirschenman, W., Heese, H.S., King, R., McConnell, B. (2024). “Automated Vessel Selection and Combat Load Planning.” Presentation at the Annual Meeting of the Institute for Operations Research and the Management Sciences (INFORMS), Seattle, WA. (Link)

Kirschenman, W., Heese, H.S., King, R., McConnell, B. (2024). “Automated Vessel Selection and Combat Load Planning.” Presentation at the 2024 Army Operations Research Symposium (AORS) Symposium, Aberdeen Proving Ground, MD. (Link)

Kirschenman, W., Brown, K. (2022). “Distributed Wargaming Visualization and Data Collection.” Presentation at the 90th Military Operations Research Society (MORS) Symposium, Marine Corps Base Quantico, VA. (Link)

Kirschenman, W., Brown, K. (2021). “Distributed Wargaming Visualization and Data Collection.” Presentation at the 2021 Army Operations Research Symposium (Virtual). (Link)

TEACHING AND ACADEMIC SERVICE

Invited Instructor
Incoming ISE/OR PhD Student Bootcamp

August 7, 2025
North Carolina State University

- Selected to design and teach a one-day “Introduction to Python” module for all incoming Ph.D. students in the Department of Industrial and Systems Engineering. (Link)

Panelist
Navigating a Ph.D., College of Engineering Smooth Sailing Series

2025 (Numerous)
North Carolina State University

- Nominated by the department to serve on a university-wide panel advising new engineering Ph.D. students. Participated in all iterations.

TECHNICAL SKILLS

Programming Languages: Julia, Python, R, SQL

Developer Tools & Version Control: Git, Subversion (SVN)

AI Development: Agentic AI & Coding, Deep Learning (Computer Vision), Machine Learning; Proficient with AI coding assistants (Claude Code, Codex, Cursor) and libraries such as PyTorch, CUDA, and ROCm.

Scientific Computing & Optimization: Mathematical Programming (exact methods, mathheuristics, complex solver callbacks, warm-starting, solver parameter tuning), High-Performance Computing (HPC), Distributed & Parallel Computing (CPU/GPU), Bayesian Inference; Proficient with Gurobi, CPLEX, and R Shiny.

Software Paradigms: Collaborative Software Development, Object-Oriented Programming (OOP)

PROFESSIONAL EXPERIENCE

The Research and Analysis Center (TRAC), Army Futures Command (AFC)
Combat Operations Analyst

August 2020 - May 2023
White Sands Missile Range, NM

- Led the development of TRAC’s Visualization Application with Data collection in R (VADR), a cloud-based planning and analytical tool for wargaming and other data collection events.
- Led planning, execution and analysis for AFC’s 2020 Capstone Wargame focused on the U.S. European Command area of responsibility.
- Planned and executed warfighter touchpoints, operational unit focus groups, and a strategic commanders assessment for AFC’s Project Convergence 22, a Combined Joint Force experiment.

2nd Infantry Brigade Combat Team, 4th Infantry Division
Company Commander, Brigade Engineer Plans Officer

August 2015 - December 2018
Fort Carson, CO

- Company Commander:** Commanded two distinct Engineer companies of over 100 Soldiers each, responsible for all training, morale, and equipment valued in excess of \$35 million.

- **Brigade Engineer Plans Officer:** Served as the primary engineer planner for a 3,500-Soldier Brigade Combat Team and as a plans officer for Combined Joint Engineer operations in southern Afghanistan.

G-3 Engineer Section, 2nd Infantry Division

Operations Officer

July 2014 - July 2015

Camp Red Cloud, Republic of Korea

- Planned, synchronized, and assessed engineer operations for the ROK–U.S. Combined 2nd Infantry Division.

Security Force Advise/Assist Team (SFAAT) #2

Engineer Company Commander Advisor

December 2012 - September 2013

Logar Province, Afghanistan

- Advised an Afghan National Army (ANA) Engineer Company Commander over nine months in all areas of command.

11th Engineer Battalion, 20th Engineer Brigade

Company Executive Officer, Platoon Leader

February 2011 - December 2013

Fort Benning, GA

- **Company Executive Officer:** Managed operations, training, and maintenance processes for a 198-Soldier Engineer Company.
- **Platoon Leader:** Led a 64-Soldier Engineer Platoon responsible for equipment valued in excess of \$20 million.

PROFESSIONAL MEMBERSHIPS

Military Operations Research Society (MORS)

2021 - Present

The Institute for Operations Research and the Management Sciences (INFORMS)

2023 - Present

Engineer Intern in New York

2010

SELECTED MILITARY AWARDS

Bronze Star Medal (2), Meritorious Service Medal (2), Army Commendation Medal (2), Army Achievement Medal (5), Military Outstanding Volunteer Service Medal, Steel de Fleury Medal (Army Engineer Association)

COMMUNITY SERVICE

Cub Scout Leader, White Sands Missile Range, NM

2021 - 2022

Pianist for Cascade Hills Church (GA) and Joyful Baptist Church (CO)

2011 - 2018